

Lesson 3: Curvilinear motion in rectangular coordinates

Lesson 2: Curvilinear motion in rectangular coordinates

1. A drone is flying in the xy plane. The operator of the drone is located at the origin, O . The motion of the drone is defined by the position vector

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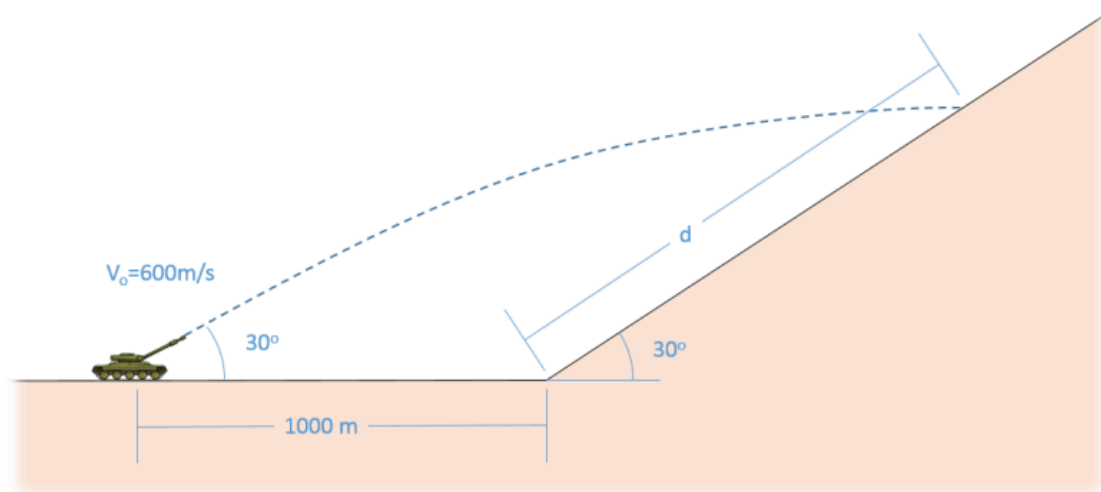
$$\vec{r} = (3 + 2t^3)\hat{i} + (3t^2)\hat{j}$$

here r and t are expressed in meters and seconds, respectively. Determine (a) the distance between the drone and its operator at $t=2$, (b) the magnitude of the drone's velocity at $t=2$ s. (c) the magnitude of the drone's acceleration at $t=2$ s.

2. A shot put is released from a height of 1.8 m with an initial speed of 12 m/s at a 40° angle above the horizontal. How far, in meters, does the shot travel?

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3. A tank fires a round at a 30° angle with a muzzle velocity of 600 m/s. The round is expected to hit a mountainside one kilometer away. The mountainside also has an average angle of 30° . How far up the mountainside, in kilometers, will the round be expected to travel before hitting the ground, labeled d in the image, if we ignore air resistance? /1



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4. An archer fires an arrow at a target that is 100 yards away. The target center is 60 inches off the ground. The arrow is released at a height of 72 inches from the ground with a speed of 300 fps. What firing angle, in degrees, is required to hit the target? /1